

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM2.5 (pm25)
Observed / expected baseline	41.2 / 8.22963 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-04 16:00 UTC
Location	29.6958, -95.4992
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 637	60.39 to 100; mean 87.3765	88.78 at 2026-06-04 15:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 637	21.6667 to 31.1111; mean 25.4696	26 at 2026-06-04 15:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 636	0 to 360; mean 97.9717	70 at 2026-06-04 15:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 637	0 to 9.77444; mean 3.2926	3.08666 at 2026-06-04 15:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 120	5862.22 to 5884.53; mean 5874.15	5878.94 at 2026-06-04 18:00Z; 2 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 120	33.5323 to 1330.69; mean 475.451	1288.53 at 2026-06-04 18:00Z; 2 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 120	42.8617 to 56.6819; mean 52.1256	50.8687 at 2026-06-04 18:00Z; 2 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 120	1003.6 to 1015.16; mean 1010.99	1011.66 at 2026-06-04 18:00Z; 2 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 120	16.13 to 19.4519; mean 17.1357	16.7238 at 2026-06-04 18:00Z; 2 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 120	-4.29892 to -0.40627; mean -2.5509	-3.46289 at 2026-06-04 18:00Z; 2 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 120	-3.20242 to 3.14362; mean 0.3119	-0.0512231 at 2026-06-04 18:00Z; 2 h after event
openaq	Ozone (ozone)	ppm	13 / 928	0 to 0.046; mean 0.0195	0.016 at 2026-06-04 16:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	2 / 144	6 to 39; mean 17.1806	16 at 2026-06-04 16:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	27 / 2922	-4 to 42.5; mean 5.5579	41.2 at 2026-06-04 16:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 78.1083	69 at 2026-06-04 15:57Z; 2 min before event
openweather	Relative humidity (humidity)	percent	5 / 360	66 to 97; mean 87.2194	84 at 2026-06-04 15:57Z; 2 min before event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 21.77; mean 0.41	0 at 2026-06-04 15:57Z; 2 min before event
openweather	Surface pressure (pressure)	hPa	5 / 360	1010 to 1017; mean 1014.07	1016 at 2026-06-04 15:57Z; 2 min before event
openweather	Air temperature (temperature)	degC	5 / 360	22.15 to 30.99; mean 25.6216	27.12 at 2026-06-04 15:57Z; 2 min before event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 355; mean 103.997	238 at 2026-06-04 15:57Z; 2 min before event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.45 to 6.28; mean 2.2846	0.45 at 2026-06-04 15:57Z; 2 min before event
purpleair	PM2.5 (pm25)	ug/m3	63 / 4585	0 to 40.4; mean 5.9539	15.8 at 2026-06-04 16:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	3 / 3	0.0277457 to 0.0294826; mean 0.0283	0.0277806 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	4 / 4	8.2537e-05 to 0.000140136; mean 0.0001	0.000135539 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	2 / 2	4.64928e-05 to 4.7738e-05; mean 0	4.64928e-05 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	4 / 4	-9.61344e-06 to 1.0299e-05; mean 0	1.55388e-06 at 2026-06-04 18:45Z; 2 h 45 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-04 18:45Z; 2 h 45 min after event
tceq	Carbon monoxide (co)	ppm	3 / 214	0 to 0.7; mean 0.185	0.1 at 2026-06-04 16:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	13 / 913	-2.6 to 28.3; mean 7.2315	11 at 2026-06-04 16:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 216	-0.6 to 3.6; mean -0.0356	-0.2 at 2026-06-04 16:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-03 00Z 86.7, 06Z 89.65, 12Z 79.25, 18Z 90.35, 06-04 00Z 93.09, 06Z 95.89, 12Z 87.89, 18Z 77.08, 06-05 00Z 87.68, 06Z 93.46, 12Z 87.88, 18Z 81.6, 06-06 00Z 84.35
asos	Air temperature (temperature)	06-03 00Z 23.74, 06Z 23.61, 12Z 26.51, 18Z 24.58, 06-04 00Z 23.74, 06Z 23.16, 12Z 25.41, 18Z 27.81, 06-05 00Z 25.74, 06Z 24.84, 12Z 26.49, 18Z 27.9, 06-06 00Z 26.57
asos	Wind direction (wind_direction)	06-03 00Z 105.3, 06Z 149.6, 12Z 75.85, 18Z 80.75, 06-04 00Z 70, 06Z 72.69, 12Z 67.22, 18Z 113.3, 06-05 00Z 103, 06Z 100.7, 12Z 116, 18Z 114.4, 06-06 00Z 117.5
asos	Wind speed (wind_speed)	06-03 00Z 3.807, 06Z 3.077, 12Z 3.296, 18Z 2.883, 06-04 00Z 2.734, 06Z 1.89, 12Z 2.401, 18Z 5.144, 06-05 00Z 3.145, 06Z 3.153, 12Z 3.354, 18Z 4.64, 06-06 00Z 3.73
noaa_gfs	500-hPa geopotential height (gh_500)	06-03 06Z 5880, 12Z 5869, 18Z 5878, 06-04 00Z 5868, 06Z 5875, 12Z 5869, 18Z 5879, 06-05 00Z 5876, 06Z 5884, 12Z 5871, 18Z 5876, 06-06 00Z 5864
noaa_gfs	Planetary boundary layer height (pbl_height)	06-03 06Z 217.3, 12Z 182.7, 18Z 665.5, 06-04 00Z 346.3, 06Z 143.5, 12Z 122.7, 18Z 1111, 06-05 00Z 673, 06Z 354.9, 12Z 238.7, 18Z 952.6, 06-06 00Z 697.3
noaa_gfs	Precipitable water (precipitable_water)	06-03 06Z 48.36, 12Z 51.92, 18Z 53.05, 06-04 00Z 51.19, 06Z 50.24, 12Z 49.87, 18Z 51.82, 06-05 00Z 54.8, 06Z 52.31, 12Z 53.96, 18Z 55.28, 06-06 00Z 52.71
noaa_gfs	Surface pressure (surface_pressure)	06-03 06Z 1012, 12Z 1013, 18Z 1013, 06-04 00Z 1011, 06Z 1012, 12Z 1012, 18Z 1012, 06-05 00Z 1010, 06Z 1010, 12Z 1010, 18Z 1009, 06-06 00Z 1007
noaa_gfs	850-hPa temperature (t_850)	06-03 06Z 18.97, 12Z 16.65, 18Z 16.33, 06-04 00Z 16.95, 06Z 16.9, 12Z 16.45, 18Z 16.61, 06-05 00Z 17.11, 06Z 17.12, 12Z 17, 18Z 17.4, 06-06 00Z 18.13
noaa_gfs	10-m eastward wind (u_10m)	06-03 06Z -2.493, 12Z -1.226, 18Z -3.109, 06-04 00Z -2.441, 06Z -1.849, 12Z -1.49, 18Z -3.216, 06-05 00Z -3.568, 06Z -2.686, 12Z -2.444, 18Z -2.491, 06-06 00Z -3.596
noaa_gfs	10-m northward wind (v_10m)	06-03 06Z 0.4092, 12Z -2.37, 18Z -0.8602, 06-04 00Z 0.02796, 06Z -0.3783, 12Z -0.8734, 18Z -0.1282, 06-05 00Z 2.119, 06Z 1.097, 12Z 0.1544, 18Z 2.227, 06-06 00Z 2.319
openaq	Ozone (ozone)	06-03 00Z 0.03291, 06Z 0.02039, 12Z 0.02021, 18Z 0.02169, 06-04 00Z 0.01297, 06Z 0.009936, 12Z 0.01128, 18Z 0.02563, 06-05 00Z 0.01963, 06Z 0.02246, 12Z 0.02074, 18Z 0.02526, 06-06 00Z 0.01952
openaq	PM10 (pm10)	06-03 00Z 10.75, 06Z 15.83, 12Z 22.08, 18Z 15.27, 06-04 00Z 18.33, 06Z 20.67, 12Z 15.08, 18Z 13.17, 06-05 00Z 21.83, 06Z 19.5, 12Z 14, 18Z 11.91, 06-06 00Z 20.6
openaq	PM2.5 (pm25)	06-03 00Z 4.421, 06Z 4.94, 12Z 7.158, 18Z 5.391, 06-04 00Z 6.651, 06Z 8.346, 12Z 7.873, 18Z 4.111, 06-05 00Z 4.924, 06Z 5.534, 12Z 5.241, 18Z 3.807, 06-06 00Z 2.518
openweather	Cloud cover (cloud_cover)	06-03 00Z 62.6, 06Z 89.53, 12Z 95.43, 18Z 84.5, 06-04 00Z 97.4, 06Z 97.8, 12Z 72.13, 18Z 44.77, 06-05 00Z 38, 06Z 48.1, 12Z 82.73, 18Z 99.37, 06-06 00Z 100
openweather	Relative humidity (humidity)	06-03 00Z 85.8, 06Z 87.47, 12Z 82.67, 18Z 83.73, 06-04 00Z 92.47, 06Z 93.8, 12Z 88.7, 18Z 79.1, 06-05 00Z 86.9, 06Z 92.6, 12Z 88.23, 18Z 84.93, 06-06 00Z 86.15
openweather	Precipitation rate (precipitation)	06-03 00Z 0, 06Z 0, 12Z 0.5023, 18Z 1.794, 06-04 00Z 0, 06Z 0.05033, 12Z 0.3743, 18Z 0.08, 06-05 00Z 0, 06Z 0.4833, 12Z 1.18, 18Z 0.4553, 06-06 00Z 0
openweather	Surface pressure (pressure)	06-03 00Z 1015, 06Z 1015, 12Z 1017, 18Z 1015, 06-04 00Z 1015, 06Z 1015, 12Z 1016, 18Z 1013, 06-05 00Z 1014, 06Z 1013, 12Z 1013, 18Z 1011, 06-06 00Z 1011
openweather	Air temperature (temperature)	06-03 00Z 23.68, 06Z 23.55, 12Z 26.61, 18Z 25.7, 06-04 00Z 23.57, 06Z 22.99, 12Z 25.59, 18Z 28.5, 06-05 00Z 25.79, 06Z 24.8, 12Z 26.61, 18Z 28.08, 06-06 00Z 26.67
openweather	Wind direction (wind_direction)	06-03 00Z 111.2, 06Z 93.13, 12Z 55.87, 18Z 136.4, 06-04 00Z 88.87, 06Z 108.6, 12Z 95.1, 18Z 119.5, 06-05 00Z 107.1, 06Z 89.3, 12Z 120.8, 18Z 116.5, 06-06 00Z 119.7
openweather	Wind speed (wind_speed)	06-03 00Z 1.72, 06Z 1.638, 12Z 2.364, 18Z 4.105, 06-04 00Z 1.922, 06Z 1.32, 12Z 2.212, 18Z 2.773, 06-05 00Z 2.201, 06Z 1.892, 12Z 2.308, 18Z 2.533, 06-06 00Z 2.361

Source	Metric	Values
purpleair	PM2.5 (pm25)	06-03 00Z 4.746, 06Z 6.599, 12Z 8.375, 18Z 5.736, 06-04 00Z 8.226, 06Z 11.32, 12Z 8.619, 18Z 3.333, 06-05 00Z 5.251, 06Z 5.468, 12Z 3.856, 18Z 2.426, 06-06 00Z 2.047
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-03 18Z 1, 06-04 18Z 1, 06-05 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-03 18Z 1, 06-04 18Z 1, 06-05 18Z 1
sentinel5p	CO column (s5p_co_column)	06-03 18Z 0.02948, 06-04 18Z 0.02776
sentinel5p	CO granules available (s5p_co_granule_available)	06-03 18Z 1, 06-04 18Z 1, 06-05 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-04 18Z 0.0001378, 06-05 18Z 8.291e-05
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-03 18Z 1, 06-04 18Z 1, 06-05 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-03 18Z 1, 06-04 18Z 1, 06-05 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-03 18Z 1, 06-04 18Z 1, 06-05 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-04 18Z -4.03e-06, 06-05 18Z 5.139e-06
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-03 18Z 1, 06-04 18Z 1, 06-05 18Z 1
tceq	Carbon monoxide (co)	06-03 00Z 0.1167, 06Z 0.1222, 12Z 0.1667, 18Z 0.2, 06-04 00Z 0.1611, 06Z 0.1444, 12Z 0.1882, 18Z 0.1889, 06-05 00Z 0.2333, 06Z 0.15, 12Z 0.2125, 18Z 0.2, 06-06 00Z 0.3
tceq	Nitrogen dioxide (no2)	06-03 00Z 5.183, 06Z 6.299, 12Z 8.659, 18Z 12.24, 06-04 00Z 9.399, 06Z 8.38, 12Z 9.249, 18Z 4.309, 06-05 00Z 5.353, 06Z 5.053, 12Z 7.146, 18Z 5.435, 06-06 00Z 5.391
tceq	Sulfur dioxide (so2)	06-03 00Z -0.01667, 06Z -0.02778, 12Z 0.55, 18Z 0.1313, 06-04 00Z -0.2056, 06Z -0.2556, 12Z -0.1529, 18Z -0.02778, 06-05 00Z -0.06667, 06Z -0.1333, 12Z -0.1167, 18Z -0.07222, 06-06 00Z -0.04667

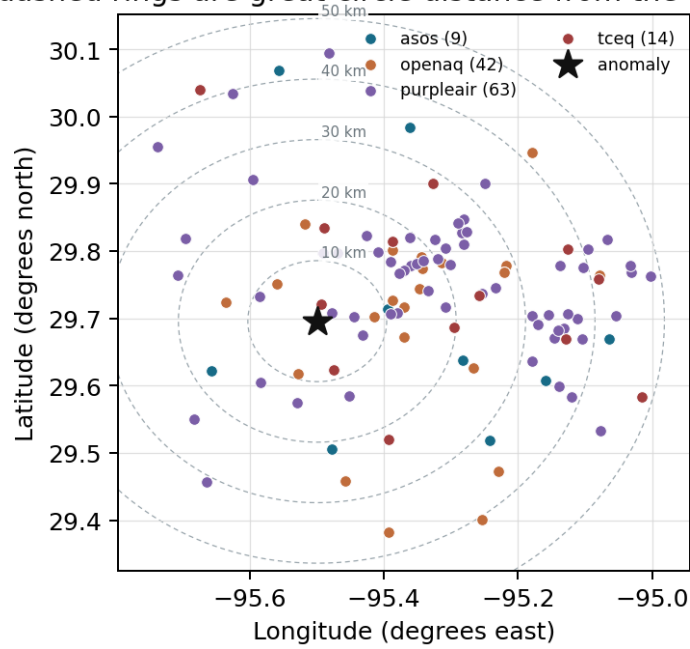
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

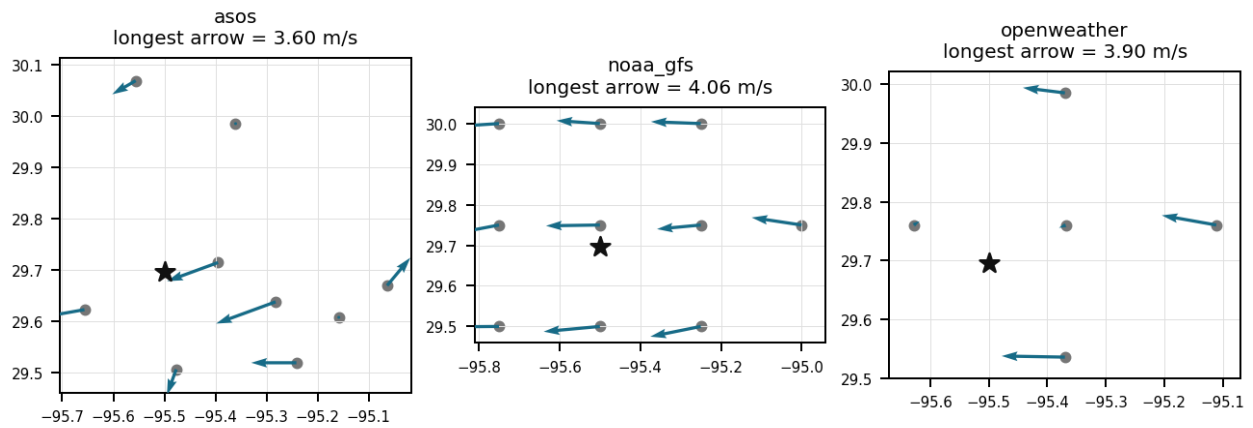
Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (10.265 km) 90.1, SGR (17.257 km) 86.86, AXH (21.204 km) 80.78, HOU (21.927 km) 85.97, LVJ (31.728 km) 86.2, EFD (34.344 km) 89.63, IAH (34.76 km) 90.58, DWH (41.75 km) 88.77, +1 more
asos	Air temperature (temperature)	MCJ (10.265 km) 25.06, SGR (17.257 km) 25.77, AXH (21.204 km) 25.98, HOU (21.927 km) 25.73, LVJ (31.728 km) 25.82, EFD (34.344 km) 25.71, IAH (34.76 km) 24.7, DWH (41.75 km) 24.71, +1 more
asos	Wind direction (wind_direction)	MCJ (10.265 km) 107.9, SGR (17.257 km) 102.1, AXH (21.204 km) 87.36, HOU (21.927 km) 103.6, LVJ (31.728 km) 107.5, EFD (34.344 km) 88.1, IAH (34.76 km) 107.5, DWH (41.75 km) 77.36, +1 more
asos	Wind speed (wind_speed)	MCJ (10.265 km) 4.308, SGR (17.257 km) 3.123, AXH (21.204 km) 1.979, HOU (21.927 km) 3.551, LVJ (31.728 km) 2.722, EFD (34.344 km) 3.83, IAH (34.76 km) 3.515, DWH (41.75 km) 2.579, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.50 (6.027 km) 5874, gfs:29.50,-95.50 (21.772 km) 5875, gfs:29.75,-95.25 (24.807 km) 5874, gfs:29.75,-95.75 (24.957 km) 5874, gfs:29.50,-95.25 (32.474 km) 5875, gfs:29.50,-95.75 (32.589 km) 5875, gfs:30.00,-95.50 (33.826 km) 5874, gfs:30.00,-95.25 (41.495 km) 5874, +2 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.50 (6.027 km) 536.4, gfs:29.50,-95.50 (21.772 km) 455.5, gfs:29.75,-95.25 (24.807 km) 493.2, gfs:29.75,-95.75 (24.957 km) 451.6, gfs:29.50,-95.25 (32.474 km) 458.9, gfs:29.50,-95.75 (32.589 km) 439.7, gfs:30.00,-95.50 (33.826 km) 521.4, gfs:30.00,-95.25 (41.495 km) 546.1, +2 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.50 (6.027 km) 52.09, gfs:29.50,-95.50 (21.772 km) 52.26, gfs:29.75,-95.25 (24.807 km) 52.74, gfs:29.75,-95.75 (24.957 km) 51.6, gfs:29.50,-95.25 (32.474 km) 52.61, gfs:29.50,-95.75 (32.589 km) 51.79, gfs:30.00,-95.50 (33.826 km) 51.66, gfs:30.00,-95.25 (41.495 km) 52.24, +2 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.50 (6.027 km) 1011, gfs:29.50,-95.50 (21.772 km) 1012, gfs:29.75,-95.25 (24.807 km) 1012, gfs:29.75,-95.75 (24.957 km) 1010, gfs:29.50,-95.25 (32.474 km) 1012, gfs:29.50,-95.75 (32.589 km) 1011, gfs:30.00,-95.50 (33.826 km) 1010, gfs:30.00,-95.25 (41.495 km) 1011, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.50 (6.027 km) 17.18, gfs:29.50,-95.50 (21.772 km) 17.17, gfs:29.75,-95.25 (24.807 km) 17.14, gfs:29.75,-95.75 (24.957 km) 17.14, gfs:29.50,-95.25 (32.474 km) 17.11, gfs:29.50,-95.75 (32.589 km) 17.25, gfs:30.00,-95.50 (33.826 km) 17.16, gfs:30.00,-95.25 (41.495 km) 17.06, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.50 (6.027 km) -2.415, gfs:29.50,-95.50 (21.772 km) -2.69, gfs:29.75,-95.25 (24.807 km) -2.51, gfs:29.75,-95.75 (24.957 km) -2.336, gfs:29.50,-95.25 (32.474 km) -2.775, gfs:29.50,-95.75 (32.589 km) -2.66, gfs:30.00,-95.50 (33.826 km) -2.45, gfs:30.00,-95.25 (41.495 km) -2.534, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.50 (6.027 km) 0.2935, gfs:29.50,-95.50 (21.772 km) 0.2076, gfs:29.75,-95.25 (24.807 km) 0.2843, gfs:29.75,-95.75 (24.957 km) 0.3068, gfs:29.50,-95.25 (32.474 km) 0.446, gfs:29.50,-95.75 (32.589 km) 0.2451, gfs:30.00,-95.50 (33.826 km) 0.2126, gfs:30.00,-95.25 (41.495 km) 0.216, +2 more
openaq	Ozone (ozone)	311 (0.0 km) 0.02011, 312 (8.352 km) 0.01872, 287 (13.542 km) 0.02253, 319 (15.42 km) 0.01596, 325 (19.779 km) 0.0171, 3378 (23.723 km) 0.01656, 320 (23.737 km) 0.01862, 2046 (25.439 km) 0.01637, +5 more
openaq	PM10 (pm10)	13380082 (23.723 km) 18.17, 271 (35.925 km) 16.19
openaq	PM2.5 (pm25)	2071339 (0.0 km) 7.847, 8967001 (8.221 km) 5.475, 14140748 (8.437 km) 5.357, 15399615 (9.045 km) 4.762, 12286630 (11.357 km) 3.84, 16123962 (12.759 km) 4.567, 9416627 (12.832 km) 4.144, 2071333 (13.542 km) 0.1875, +19 more
openweather	Cloud cover (cloud_cover)	grid:west (14.404 km) 76.4, grid:center (14.533 km) 76.01, grid:south (21.749 km) 81.33, grid:north (34.495 km) 78.68, grid:east (38.157 km) 78.11
openweather	Relative humidity (humidity)	grid:west (14.404 km) 86, grid:center (14.533 km) 85.97, grid:south (21.749 km) 84.51, grid:north (34.495 km) 89.07, grid:east (38.157 km) 90.54
openweather	Precipitation rate (precipitation)	grid:west (14.404 km) 0.1424, grid:center (14.533 km) 0.3121, grid:south (21.749 km) 0.2465, grid:north (34.495 km) 0.5889, grid:east (38.157 km) 0.76
openweather	Surface pressure (pressure)	grid:west (14.404 km) 1014, grid:center (14.533 km) 1014, grid:south (21.749 km) 1014, grid:north (34.495 km) 1014, grid:east (38.157 km) 1014
openweather	Air temperature (temperature)	grid:west (14.404 km) 25.64, grid:center (14.533 km) 25.64, grid:south (21.749 km) 26, grid:north (34.495 km) 25.04, grid:east (38.157 km) 25.79
openweather	Wind direction (wind_direction)	grid:west (14.404 km) 124.2, grid:center (14.533 km) 108.4, grid:south (21.749 km) 104.1, grid:north (34.495 km) 83.19, grid:east (38.157 km) 100
openweather	Wind speed (wind_speed)	grid:west (14.404 km) 1.518, grid:center (14.533 km) 1.844, grid:south (21.749 km) 2.44, grid:north (34.495 km) 2.234, grid:east (38.157 km) 3.387
purpleair	PM2.5 (pm25)	186315 (2.571 km) 8.104, 46237 (5.403 km) 0.1329, 166451 (6.9 km) 8.007, 28851 (9.308 km) 7.638, 293735 (10.609 km) 8.468, 51959 (11.328 km) 6.11, 293725 (11.534 km) 8.679, 301969 (11.71 km) 8.981, +55 more
tceq	Carbon monoxide (co)	482011052 (17.015 km) 0.2681, 482011035 (23.71 km) 0.1681, 482011039 (35.926 km) 0.1192
tceq	Nitrogen dioxide (no2)	482010055 (0.006 km) 7.632, 482011066 (2.94 km) 10.52, 482010047 (15.421 km) 10.94, 482011052 (17.015 km) 11.19, 482010416 (19.779 km) 10.38, 480391004 (22.058 km) 5.405, 482011035 (23.71 km) 9.086, 482010024 (28.279 km) 0.02889, +5 more
tceq	Sulfur dioxide (so2)	482010051 (8.339 km) -0.2071, 482011035 (23.71 km) -0.1726, 482011039 (35.926 km) 0.2658

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO₂ particles.

Mark exactly one:

☐ V

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Note (optional):

Claim 2 of 36

A localized combustion-related fine-particle plume near the Houston site is a plausible cause because PM_{2.5} reached 41.2 ug/m³ at the anomaly site while nearby PM₁₀ was only 16.0 ug/m³, ozone at the site was only 0.016 ppm, TCEQ SO₂ was near zero, and regional satellite CO was not elevated, which fits a short-lived local primary PM_{2.5} event more than a broad regional photochemical or sulfur-rich plume.

Mark exactly one:

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Note (optional):

Claim 3 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:

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Note (optional):

Claim 4 of 36

The openaq PM2.5 measurement at 29.696, -95.499 was 41.2 ug/m3 at 2026-06-04T16:00:00+00:00, compared with an expected value of 8.23 ug/m3, corresponding to a z-score of 10.50 and severe anomaly classification.

Mark exactly one:

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Note (optional):

Claim 5 of 36

The anomaly is a result of a combination of factors including temperature inversions, atmospheric stagnation, and increased emissions from vehicles and industrial activities in the Houston area.

Mark exactly one:

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Note (optional):

Claim 6 of 36

High relative humidity levels exceeding 85% near the anomaly location facilitated hygroscopic aerosol growth and particulate accumulation.

Mark exactly one:

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Note (optional):

Claim 7 of 36

Weak to modest and somewhat variable near-surface winds likely favored temporary accumulation of a neighborhood-scale plume because local wind speeds near the anomaly were low in some observations, including about 0.45 m/s in openweather and about 3.09 m/s at the nearest ASOS station, while wind directions differed between datasets, which is consistent with limited dispersion and possible microscale recirculation.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.5007214681470114 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 8 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:

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Note (optional):

Claim 9 of 36

A nearby PurpleAir sensor located 2.571 km from the anomaly site recorded elevated PM2.5 levels of 15.8 ug/m3 at 16:00 UTC on June 4, 2026.

Mark exactly one:

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Note (optional):

Claim 10 of 36

The location of the anomaly is approximately 2.571 km from the nearest sensor.

Mark exactly one:

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Note (optional):

Claim 11 of 36

High ambient relative humidity exceeding 84 percent observed by ASOS and OpenWeather promoted aerosol hygroscopic growth and secondary particle formation, contributing to elevated PM2.5 readings detected by OpenAQ.

Mark exactly one:

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Note (optional):

Claim 12 of 36

Humidity likely amplified the measured Houston PM2.5 anomaly because relative humidity near the anomaly was very high at about 84 to 89 percent around 16:00 UTC, but humidity alone is not sufficient to explain the event because the local PM2.5 spike was far larger than the area-wide 6-hour PM2.5 means and occurred when modeled afternoon boundary-layer height was high near 1289 m, which would usually reduce accumulation from stagnant trapping.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 13 of 36**

The Houston PM2.5 anomaly is strongly consistent with a localized fine-particle plume because the anomalous openaq PM2.5 value reached 41.2 ug/m3 while a nearby purpleair sensor at the same hour measured only 15.8 ug/m3 and the nearest openaq PM10 value was only 16.0 ug/m3, indicating sharp spatial gradients and a fine-mode-dominant signal rather than a uniform regional event.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 14 of 36**

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 15 of 36

The air quality anomaly in Houston, Texas on June 4th was likely caused by a combination of factors including weather patterns and local emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 16 of 36**

The air quality anomaly in Houston, Texas on June 4th was also characterized by elevated levels of NO₂.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 17 of 36**

The openaq PM_{2.5} anomaly at the Houston site is most consistent with a localized primary fine-particle plume rather than a region-wide PM_{2.5} episode.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 18 of 36**

Light surface winds under 3 m/s and a shallow boundary layer trapped local primary combustion emissions near the surface, as indicated by elevated TCEQ NO₂ concentrations.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 36

The anomaly occurred on June 4th, 2026 at 16:00:00+00:00.

Mark exactly one:

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Note (optional):

Claim 20 of 36

Stagnant meteorological conditions characterized by low surface wind speeds and a shallow morning planetary boundary layer restricted vertical mixing and trapped fine particulate matter near the ground.

Mark exactly one:

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Note (optional):

Claim 21 of 36

OpenAQ recorded a PM2.5 concentration spike to 41.2 ug/m3 at 16:00 UTC on June 4, 2026 in Houston, Texas.

Mark exactly one:

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Note (optional):

Claim 22 of 36

Boundary layer suppression and weak surface winds facilitated localized particulate entrapment, with GFS showing morning planetary boundary layer heights dropping to near 122 meters and OpenWeather reporting surface wind speeds down to 0.45 m/s near the time of the event.

Mark exactly one:

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Note (optional):

Claim 23 of 36

High ambient relative humidity exceeding 85 percent promoted the hygroscopic growth of aerosol particles and aqueous-phase conversion of precursor gases into secondary fine particulate matter.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 24 of 36**

High ambient humidity likely amplified the observed PM_{2.5} concentration through hygroscopic growth or measurement bias because relative humidity near the anomaly time was about 84 to 89 percent across nearby observations, conditions under which fine particles can take up water and increase reported PM_{2.5} mass.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 25 of 36**

At 16:00 UTC on June 4, 2026, a TCEQ monitor situated 0.006 km from the anomaly recorded a nitrogen dioxide (NO₂) concentration of 11.0 ppb.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 26 of 36

The air quality anomaly in Houston, Texas on June 4th was characterized by elevated levels of PM2.5.

Mark exactly one:

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Note (optional):

Claim 27 of 36

On June 4, 2026, at 16:00 UTC, an OpenAQ surface monitor at coordinates (29.696, -95.499) in Houston, Texas, recorded a PM2.5 spike to 41.2 ug/m3, well above the expected value of 8.23 ug/m3.

Mark exactly one:

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Note (optional):

Claim 28 of 36

A broad regional secondary photochemical or sulfur-rich pollution episode is weakly supported because ozone near the anomaly was low at 0.016 ppm, TCEQ SO2 near the anomaly was near zero or negative, Sentinel-5P SO2 was near zero, and Sentinel-5P CO on 2026-06-04 was not elevated relative to 2026-06-03, although elevated Sentinel-5P HCHO leaves open some contribution from reactive VOC-rich combustion or industrial precursors.

Mark exactly one:

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Note (optional):

Claim 29 of 36

High ambient humidity in Houston likely increased the measured PM_{2.5} during the plume through particle-water uptake, but high humidity alone is insufficient to explain the full magnitude of the PM_{2.5} spike.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.5007214681470114 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 30 of 36

Localized combustion emissions from urban transport or industrial activities contributed primary aerosols and precursors, supported by elevated co-located nitrogen dioxide readings.

Mark exactly one:

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Note (optional):

Claim 31 of 36

A severe thunderstorm or tornado event has stirred up dust and pollutants from the ground, contributing to the air quality anomaly.

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Note (optional):

Claim 32 of 36

At 2026-06-04T16:00:00+00:00 near 29.696, -95.499, openaq ozone was 0.016 ppm at the co-located sensor and openaq PM10 was 16.0 ug/m3 at the nearest PM10 sensor 23.723 km away.

Mark exactly one:

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Note (optional):

Claim 33 of 36

Elevated local combustion emissions contributed to the PM2.5 spike, supported by co-located TCEQ nitrogen dioxide measurements reaching 11.0 ppb at the exact time and location of the OpenAQ PM2.5 peak.

Mark exactly one:

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Note (optional):

Claim 34 of 36

The nearest purpleair PM2.5 sensor, 2.571 km from 29.696, -95.499, measured 15.8 ug/m3 at 2026-06-04T16:00:00+00:00.

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Note (optional):

Claim 35 of 36

The Houston anomaly is more consistent with near-source smoke or combustion-related aerosol than with a broad photochemical, sulfur-rich, or dust-driven event.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 36 of 36**

The air quality anomaly is related to particulate matter (PM_{2.5}) with a value of 15.8 ug/m³.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.